

Maximum Units on a Truck

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Problem Overview

You are given a 2D array representation of box types: $[[a_1, b_1], [a_2, b_2], \dots]$ where:

Element	Description
a_i	Number of boxes of type i available.
b_i	Number of units contained in each box of type i .

There is a defined **truck size limit** constraint representing the maximum number of boxes the truck can carry. You can add as many boxes as you want as long as the total number of boxes does not exceed this limit.

Objective: Find the maximum possible total number of units that can be loaded onto the truck.

Greedy Observation & Strategy

To maximize the total units, we should greedily prioritize the boxes that contain the largest number of units per box. Therefore, the strategic approach is to **sort the 2D array based on the second element (b_i) in descending order**.

Trace Walkthrough Example:

Given Initial Box Types: $[[3, 1], [2, 5], [4, 3]]$

After Sorting (Descending by Units): $[[2, 5], [4, 3], [3, 1]]$

Execution Strategy: Start filling the truck sequentially from the highest yield box type ($[2, 5]$), taking as many boxes as possible until the truck size limit is perfectly reached or exhausted.

Optimal Implementation (C++)

```
class Solution {
public:
    int maximumUnits(vector<vector<int>>& boxTypes, int limit) {
        int n = boxTypes.size();
        sort(boxTypes.begin(), boxTypes.end(), [](vector<int>& a, vector<int>& b) {
            return a[1] > b[1];
        });

        int ans = 0;
        for (auto& box : boxTypes) {
            int boxes = box[0];
            int units = box[1];

            int adder = min(boxes, limit);
            ans += (adder * units);
            limit -= adder;
        }

        return ans;
    }
};
```

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